

Machine Learning vs. Deep Learning

Picking the Right Tool for the Job

Before starting this assignment I'll admit I had a fairly lazy mental model of these two ideas. Machine learning was the old thing, deep learning was the new-and-improved thing, and if you had the resources you would always want deep learning. Working through the lesson pushed me to drop that framing. Deep learning is not a straight upgrade; it is a specialist. It is extraordinary at a narrow band of problems and clumsy, wasteful, or downright worse everywhere else. The more useful question is not "which one is more advanced," it is "what does this specific problem actually need?"

To make that concrete, I chose two examples from the lesson that sit on opposite ends of that spectrum. House price prediction is a textbook machine learning problem. Image recognition is a problem that, realistically, only deep learning solves well. For each one I will point to a real system people actually use, explain why the chosen approach fits, and then explain why reaching for the other approach would be the wrong move.

Example 1 — House Price Prediction (Machine Learning)

A real system that uses it. If you have ever typed an address into Zillow and seen an estimated value pop up — the number they call the "Zestimate" — you have watched this kind of model work. Behind that estimate is an automated valuation model that takes structured details about a property (square footage, number of bedrooms and bathrooms, lot size, year built, location, recent nearby sales) and predicts a dollar figure. It is the same broad family of models that banks and professional appraisers lean on when they need a fast, defensible estimate of what a home is worth. The lesson's linear regression example is essentially the simplest honest version of this.

Why machine learning is the right fit. The thing that makes this a machine learning problem rather than a deep learning one is the shape of the data. The inputs are already clean, structured, and meaningful. "2,000 square feet" and "four bedrooms" are not raw signals a computer has to decode — they are finished features a human already understands. There is nothing hidden inside them for a model to go dig out. That plays perfectly to a technique like linear regression, which learns how much each feature nudges the price up or down and combines those effects into a single prediction.

Interpretability seals the case. Home valuation is not a casual, low-stakes task; it feeds mortgage decisions, tax assessments, and legal disputes, all of which come with real accountability. A lender needs to be able to explain and document why a valuation landed where it did, and fair-lending rules make that transparency non-negotiable. Linear regression hands you exactly that: clear, readable coefficients you can point to and say "this is why." On top of that, the amount of data involved is manageable, and a regression model trains quickly, cheaply, and reliably on it.

Why deep learning would be the wrong move. Deep neural networks are famously data-hungry; they need enormous training sets to avoid memorizing noise instead of learning real patterns. For a single housing market you simply do not have that kind of scale, so a deep model would likely overfit and

generalize worse than a plain regression. Its signature strength — teaching itself useful features from messy, raw input — is wasted here, because the features are already handed to it in clean form. There is no hierarchy of hidden structure inside “three bedrooms” for it to uncover. Worse, a deep network is a black box, and dropping a black box into a regulated, high-accountability process like lending is a genuine liability — you cannot hand a regulator or a frustrated homeowner a twelve-layer network as your explanation. In the end you would spend far more compute and complexity to get an answer that is no better, and often worse, than the simpler model. On tidy tabular data like this, the straightforward approach usually wins.

Example 2 — Image Recognition (Deep Learning)

A real system that uses it. One of the places this matters most is medicine. There are now deep learning systems that read retinal photographs to screen for diabetic retinopathy — a leading cause of blindness — and others that scan mammograms and CT images to flag possible tumors for a radiologist’s attention. These tools look at the raw medical image and pick out warning signs subtle enough to slip past a tired human eye. If a more familiar example helps, it is the same core technology behind Face ID on an iPhone, which recognizes your face from a camera image rather than from a checklist of measurements someone typed in by hand.

Why deep learning is the right fit. An image is about the least “structured” data there is. A retinal scan is millions of individual pixels, and the things that actually matter — a tiny cluster of abnormal blood vessels, a faint change in texture — cannot be boiled down to a neat list of numbers a person could write out in advance. This is precisely where deep learning earns its name. A convolutional neural network learns features in layers: the earliest layers pick up simple edges and gradients, the middle layers assemble those into textures and shapes, and the deepest layers recognize the complex, disease-specific patterns that lead to a diagnosis. Nobody has to hand-design any of that; the network discovers it from the images themselves. Convolutional networks are also built specifically for the spatial nature of pictures — they detect local patterns and recognize them no matter where in the frame they appear, which is exactly what you want when a lesion could show up anywhere on a scan. And this domain happens to have what deep learning craves: large labeled image datasets and the computing power to train on them.

Why traditional machine learning would fail here. Traditional machine learning leans on humans to define the features up front, and for images that requirement basically breaks. You would be stuck trying to hand-craft descriptors — edge maps, color histograms, and the like — and no manually designed feature set reliably captures the intricate, subtle patterns that separate a healthy retina from an early-diseased one. The feature-engineering problem is just too hard at this level of complexity. And if you skip the hand-crafting and shove raw pixels straight into something like logistic regression or a support vector machine, the model ignores all the spatial relationships that make an image meaningful and immediately drowns in dimensionality — even a small image is tens of thousands of input values with no structure it can exploit. Accuracy falls apart. These simpler models only ever learn a boundary over whatever features you feed them; they do not learn representations of their own, so weak hand-made inputs guarantee weak results. Image recognition is, honestly, the exact wall that traditional

methods kept hitting — and getting past that wall is a big part of why deep learning took off in the first place.

Conclusion

The pattern I take away from putting these two side by side is that the right choice follows from the data and the demands of the problem, not from which technique sounds more impressive. When the information is already structured and the situation calls for speed, efficiency, and an explanation you can defend, traditional machine learning is the better tool — house prices are a clean example. When the information is raw, high-dimensional, and full of patterns no human could reasonably spell out ahead of time, deep learning is the only thing that really works — images are the clear case. The lesson's own metaphor captures it nicely: sometimes you just need to hand someone a checklist, and sometimes you need to teach them to read and let them work it out for themselves. The real skill is knowing which situation you are actually in.